Across all ALD and ALE questions, we applied the 33 queries to 21 distinct model–setting configurations (for example, each model evaluated in Settings 2a, 2b, and 3), resulting in a total of 693 individual experiments. This provides a sizeable basis for comparing symbolic, neural, and symbolic-context-augmented querying.

4.1.3 Evaluation metrics

We use different evaluation metrics aligned with our three research questions.

Expert judgments on SPARQL queries (RQ1). To address RQ1, we conducted a small survey with three ALD/ALE domain experts (two of whom are co-authors of this paper). For each of the 33 queries (19 ALD, 14 ALE), experts ran the SPARQL query on the corresponding ORKG comparison using a provided ORKG link and SPARQL link, inspected the resulting table, and then rated it along two dimensions:

- *Meaningfulness*: Does the question make scientific sense (for example, does it ask about plausible relationships, use appropriate concepts, and align with current ALD/ALE knowledge)?
- *Usefulness*: Is the resulting table a helpful synthesis for researchers (for example, by saving time, surfacing patterns that are hard to see by eye, or providing non-trivial insight beyond reading the review once)?

Both dimensions were rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Experts evaluated all 33 queries spanning seven ALD and eleven ALE comparison tables. For each query, they first judged whether the natural-language question itself was meaningful, and then whether the corresponding SPARQL result table would be useful as a synthesis tool under the assumption that it could be continuously updated with additional studies in the future. Separating meaningfulness from usefulness helps diagnose whether potential improvements are needed in (i) how the question is formulated, (ii) how the underlying data are modeled in the comparison, or (iii) how the synthesized results are presented. Whenever the usefulness score was below 3, experts were asked to briefly justify their rating in a free-text field, providing qualitative feedback on limitations or missing information. The survey questionnaire is available at `https://forms.gle/qSDUG1WzSZSz2864A`; it is shared for transparency, and interested readers are welcome to inspect it or contribute their own responses.

Agreement between neural and SPARQL tables (RQ2 and RQ3). To address RQ2 and RQ3, we measure how closely the tables produced by the neural systems match the SPARQL result tables, on a per-query basis. For this, we adopt the *Relative Mapping Similarity* (RMS) metric introduced by Liu et al. [38], originally developed for evaluating table and chart question answering systems.

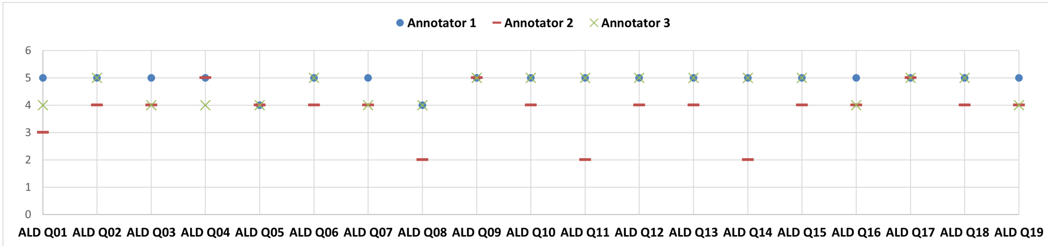
Intuitively, RMS treats each table as a set of entries of the form $(\text{row header}, \text{column header}, \text{cell value})$ and compares the predicted and gold tables by matching these entries. A high RMS score is obtained only when (i) the model selects the correct rows and columns (for example, the right materials or processes) and (ii) the corresponding values (for example, temperatures, etch rates, or EQEs) are close to those in the SPARQL table. RMS produces precision, recall, and F1 scores in the range $[0, 1]$, and is by construction insensitive to row/column order or table transposition.

In our experiments, for each query we treat the SPARQL result as the gold table and the output of each neural system (PDF-based and symbolic-context-augmented settings) as the prediction. We compute RMS precision, recall, and F1 using the official DePlot implementation from Google Research, adapted to our ALD/ALE dataset; our RMS evaluation script and configuration are documented in `https://github.com/sciknoworg/ald-ale-orkg-review/blob/main/eval_deplot_rms/README.md`. The full mathematical definition of RMS and implementation details are provided in the Appendix B.

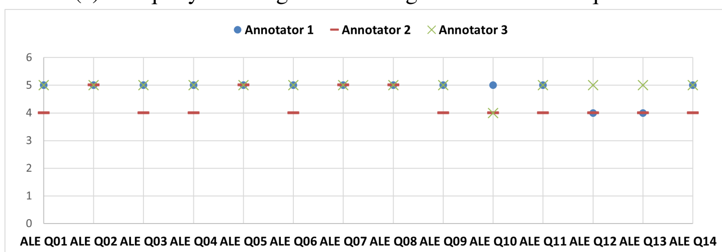
4.2 Results

4.2.1 Domain expert impressions of machine-actionable knowledge synthesis

Here we address RQ1: how useful ALD/ALE experts find ORKG-based comparison tables for answering complex review questions. As described in the Evaluation Metrics section 4.1.3, three

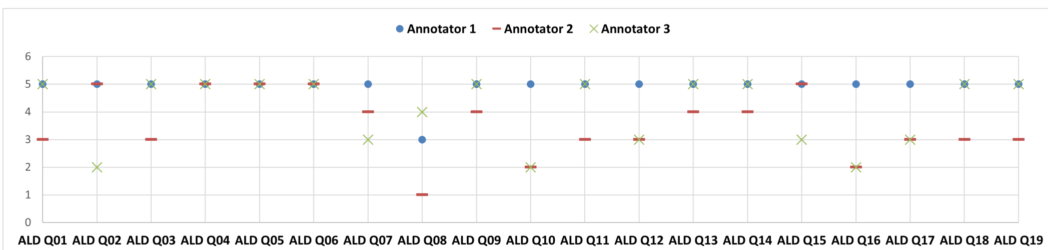


(a) Per-query meaningfulness ratings for the 19 ALD queries.

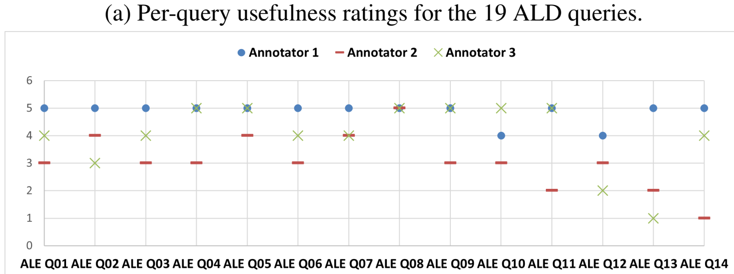


(b) Per-query meaningfulness ratings for the 14 ALE queries.

Figure 7: Meaningfulness ratings assigned by three domain experts for all ALD (a) and ALE (b) queries, on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) in response to the prompt "the query is meaningful".



(a) Per-query usefulness ratings for the 19 ALD queries.



(b) Per-query usefulness ratings for the 14 ALE queries.

Figure 8: Usefulness ratings assigned by three domain experts for all ALD (a) and ALE (b) queries, on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) in response to the prompt: "The query result provides a useful synthesis of the insights in the table" (with the instruction to imagine a richer, continuously updated version of the table when the original data are sparse).

ALD/ALE domain experts (two co-authors and one external senior researcher, with between 1–5 and > 10 years of experience in materials science) completed an online survey in which they evaluated all 33 queries (19 ALD, 14 ALE). For each query, they ran the SPARQL query on the corresponding ORKG comparison, inspected the resulting table, and rated (i) whether the question was scientifically meaningful and (ii) whether the table provided a useful synthesis.

Overall, the response was clearly positive. Experts judged almost all questions to be scientifically well-posed: the average meaningfulness score across all queries was $4.54/5$, with 96% of ratings at ≥ 4, and 32 out of 33 queries achieving an average score of at least 4. The per-query distributions of meaningfulness ratings for ALD and ALE are shown in Figure 7. Usefulness scores were slightly lower but still favourable (mean $4.08/5$), with 25 of 33 queries reaching an average usefulness of at least 4 and only two queries falling below 3; Figure 8 visualises the corresponding usefulness ratings. In a separate overall question on the value of machine-actionable scientific knowledge, all three experts rated it between 4 and 5 (mean 4.67). In other words, experts not only regard the questions as meaningful but also see the resulting SPARQL-derived tables as genuinely helpful knowledge syntheses. Detailed per-query statistics and inter-annotator agreement analyses are provided in Appendix C.

The qualitative feedback explains why. Experts repeatedly highlighted that being able to call up comparison tables "in one click" would substantially speed up literature review and make it easier to compare process conditions across many studies, especially for tasks such as precursor screening or identifying suitable process windows. They emphasised that such machine-actionable tables could support both academic and industrial work, for example by quickly surveying which precursors and co-reactants have been tried for a given material, or by mapping out temperature and plasma power ranges where acceptable growth or etch rates are reported.

At the same time, the low usefulness scores and critical remarks point to concrete areas for improvement rather than fundamental objections. Most reservations concern:

- **Missing or incomplete parameters**, such as sticking coefficients, surface orientation and termination, clearer precursor/co-reactant distinctions, dopant information, error bars on growth-per-cycle, or more detailed timing parameters per ALD/ALE cycle.
- **Schema and categorisation refinements**, including more consistent treatment of mechanisms and precursor families, and clearer separation of processes that practitioners would not class as "true" ALD/ALE.
- **Table readability**, such as avoiding duplicate rows, making very long tables more compact, and clearly exposing key identifiers (e.g., material and precursor names) rather than hiding them behind opaque resource IDs.
- **Data completeness and trust**, especially in sparsely populated comparisons or in scenarios where tables might be (partially) extracted automatically from PDFs.

The "wish-list" questions provide a particularly concrete picture of how experts would like to use such tables. Examples include querying which surface orientations and terminations are most commonly reported for a given material, grouping ALD recipes by precursor/co-reactant pair and visualising growth-per-cycle as a function of temperature, plasma power, and reactor type, or aggregating ALE conditions by material, precursor, reactant, and process temperature. These examples go beyond simple lookups and mirror the multi-parameter reasoning that ALD/ALE researchers already perform manually.

Experts also identified several factors that would affect adoption. Barriers include the additional effort required to enter machine-actionable tables alongside traditional PDFs, uneven reporting practices in the literature, concerns about quality control for automatically extracted data, and the current lack of community standards for which process and performance parameters should be captured. On the positive side, clear incentives include demonstrable time savings in literature review, community-wide schema standardisation, user-friendly tools for authoring and querying ORKG comparisons, and visible benefits for day-to-day process development (for example, being able to answer complex comparative questions "in one click").

Taken together, these findings answer RQ1 positively: ALD/ALE experts view ORKG-based machine-actionable reviews as both scientifically meaningful and practically useful. Their feedback mainly points to next steps in schema refinement, data coverage, and tooling, rather than to conceptual objections. In the remainder of this section, we therefore treat the SPARQL-derived tables as an expert-validated symbolic baseline and investigate how closely neural systems operating over PDFs and over ORKG comparisons can approximate this baseline (RQ2 and RQ3). The complete domain expert survey responses, including individual ratings and free-text comments, are available in our Zenodo repository [15].